



1. What is the value of $2^0 - (21 - 20) + 2^1$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

2. What is the smallest positive integer which can be expressed as the sum of a positive perfect square and a distinct positive perfect cube?

- (A) 2 (B) 3 (C) 5 (D) 9 (E) 10

3. Squares P and Q are such that half the area of Q lies within P . If the areas of P and Q are 8 and 2, respectively, what is the area of the region inside P and outside Q ?

- (A) 5 (B) 7 (C) 9 (D) 11 (E) 13

4. A positive integer n is divisible by 5 but not 4. For which of the following values will adding it to n never result in a sum that is divisible by 20?

- (A) 45 (B) 50 (C) 55 (D) 60 (E) 65

5. Ten points are equally spaced on the circumference of a circle, where two of the points are connected by a line segment, as shown below. Jongwook wants to choose two of the eight other points and draw a line segment connecting them so that this line segment does not intersect the other line segment. In how many ways can Jongwook do this?

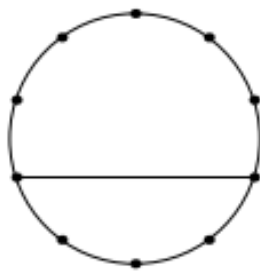


Figure 1: Problem 5

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

6. One of the side lengths of an obtuse isosceles triangle with integer side lengths is 2. What is the perimeter of this triangle?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

7. Let a be a positive integer. A geometric sequence b, c, d in that order satisfies $ab = 15$, $bd = 16$, and $ac = 120$. What is the value of $a + d$?

- (A) 56 (B) 58 (C) 60 (D) 62 (E) 64

8. Let n be the number of factors of 3 that the product $(1 + 2 + 3)(4 + 5 + 6)(7 + 8 + 9) \cdots (94 + 95 + 96)(97 + 98 + 99)$ contains. What is the sum of the digits of n ?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

9. A rectangle $ABCD$ has $AB = 8$ and $BC = 4$. Points P and Q lie on sides AB and BC , respectively, such that $AP = CQ$ and the area of $\triangle BPQ$ is 6. What is PQ^2 ?

- (A) 32 (B) 34 (C) 36 (D) 38 (E) 40

10. There exist 4-digit numbers such that the 4-digit number is divisible by 9, the first 3 digits in order form a 3-digit number divisible by 9, and the last 3 digits in order form a 3-digit number divisible by 9. How many such 4-digit numbers exist?

- (A) 16 (B) 18 (C) 20 (D) 22 (E) 24

11. In the xy -plane, the distance between the points (r, r) and $(13, 7)$ is equal to d , where r is a real number satisfying the equation $r^2 - 20r + 21 = 0$. What is d^2 ?

- (A) 78 (B) 106 (C) 150 (D) 176 (E) 218

12. Jongwook has some shirts, pairs of pants, and socks. An outfit consists of one shirt, one pair of pants, and two socks. Jongwook can currently wear 60 possible different outfits, but if he were to get either four more shirts or four more socks, then Jongwook could wear 180 possible different outfits. How many pairs of pants does Jongwook have? (The socks are distinguishable, and the order in which he wears the socks does not matter.)

- (A) 2 (B) 5 (C) 6 (D) 8 (E) 15

13. Jongwook writes the number 2021 on a blackboard. He then repeatedly erases and writes a new number on the blackboard, where if the current number on the blackboard is odd, he will erase the number and write $8n$ on the blackboard, and if the current number on the blackboard is even, he will erase the number and write $n + 2019$ on the blackboard. Eventually, Jongwook will have written 2020 numbers on the board (including the initial 2021). What is the remainder when his 2020th number is divided by 9?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 7

14. In the xy -plane, the point $(24, 7)$ is reflected over the line $y = ax$ and then shifted up b units to the point $(20, 21)$, where a and b are positive real numbers. What is $a \cdot b$?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 12

15. Jongwook repeatedly rolls two fair six-sided dice. After every roll, Jongwook writes the sum of the numbers on the two dice. Jongwook stops once he writes the number 11. What is the probability that Jongwook writes the number 5 at least once before stopping?

- (A) $\frac{1}{3}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

16. A pyramid $ABCDE$ has rectangular base $ABCD$ with $AB = 8$ and $BC = 6$ and apex E located 8 units above $ABCD$. A plane parallel to $\triangle ACE$ hits segments AB , BC , and BE at F , G , and H , respectively. If $FG = \frac{15}{2}$, what is the volume of $DFGH$?

- (A) 27 (B) 36 (C) 45 (D) 48 (E) 60

17. A positive divisor of 2021^{14} is chosen at random. What is the probability that when it is divided by 7, the remainder is equal to 1?

- (A) $\frac{1}{9}$ (B) $\frac{2}{15}$ (C) $\frac{1}{6}$ (D) $\frac{1}{5}$ (E) $\frac{1}{3}$

18. An isosceles trapezoid $ABCD$ with $AB \parallel CD$, $AB = AD = BC = 3$, and $CD = 7$ is inscribed in a circle. A point E on the circle satisfies $AC \perp BE$. What is DE^2 ?

- (A) 24 (B) 26 (C) 28 (D) 30 (E) 32

19. Let a, b, c , and d be positive integers. If $\frac{a!}{b!} + \frac{c!}{d!} = \frac{2}{5}$, what is the largest possible value of $a + b + c + d$?

- (A) 10 (B) 18 (C) 26 (D) 34 (E) 42

20. Let $\triangle ABC$ with $AB = 14$ be inscribed in a circle with center O . Let P be a point on side BC , and let line AP intersect the circle at a point D , distinct from A . If $\triangle ABC$ is acute and $ABDO$ is a rhombus, what is the largest possible integer value of BP^2 ?

- (A) 33 (B) 48 (C) 65 (D) 78 (E) 95

21. Consider the polynomial $P(x) = (x - a)(x - b)(x - c)(x - d)$, where a, b, c , and d are fixed positive real numbers less than 1. What is the maximum possible number of distinct real numbers r that can satisfy $|P(r)| = 1$?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

22. Let S be the set of all positive integers which are relatively prime to 42. What is the smallest positive integer n such that the smallest number in S with exactly 2^n positive divisors is divisible by a perfect square which is greater than 1?

- (A) 2 (B) 3 (C) 6 (D) 7 (E) 8

23. Jongwook has the string $AABBABBB$. Judo is told the string has 8 letters, each an A or a B , but not the string itself or how many of each letter are in it. Judo is told that each move, Jongwook will pick the i th letter of the existing string, that letter and every letter to its right will switch from A to B and vice versa, and Judo will learn what i is and how many of each letter are in the new string. If Jongwook moves optimally and Judo reasons well, what is the fewest number of moves for Judo to know the string?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

24. In right $\triangle ABC$ with $AB = 8$, $AC = 6$, and a right angle at A , let M be the midpoint of side BC , and let D be the reflection of C over line AM . Line segments AC and DM are extended to meet at a point E . What is the length of CE ?

(A) $\frac{32}{5}$ (B) $\frac{120}{7}$ (C) $\frac{50}{7}$ (D) $\frac{95}{13}$ (E) $\frac{84}{11}$

25. For a positive odd integer n that is not divisible by 3, consider the sequence of n numbers $1 \times 2 \times 3, 2 \times 3 \times 4, \dots, n \times (n+1) \times (n+2)$. There are 140 numbers in this sequence that are relatively prime to n . Find the sum of the digits of the largest such n .

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Time limit: 75 minutes